

AICE 2009 Lab 2 Report

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April 17, 2026

Part 1: Spectrogram of a Chirp

In this section, a linear chirp signal (100 Hz to 7000 Hz) was analyzed using the Short-Time Fourier Transform (STFT) with three different window types: Boxcar, Hann, and Hamming.

The Boxcar window produced the spectrogram with significant vertical smearing due to spectral leakage caused by its abrupt edges. The Hann window produced the smoothest and cleanest trace, as it tapers the signal perfectly to zero at the edges. The Hamming window was relatively clean but still exhibited some vertical "cuts".

The window shape determines the trade-off between the resolution of time and frequency: a longer window provides excellent frequency resolution but poor time resolution, whereas a shorter window provides exact timing but blurry frequency resolution.

Part 2: Real Audio Spectrogram

An audio file of third movement of Beethoven's Tempest sonata was analyzed using its log-magnitude spectrogram. The signal was reconstructed using the Inverse Short-Time Fourier Transform with an almost negligible reconstruction relative error. Furthermore, spectral modifications were applied using Gamma corrections ($\gamma = 0.8$ and $\gamma = 1.2$).

A Gamma factor of $\gamma = 0.8$ compresses the spectral dynamic range. Audibly, this makes the quiet background noise and subtle overtones much louder and more prominent. On the spectrogram, this compression can be seen as an overall brightening, where previously dark, low-energy regions become visible.

Conversely, a Gamma factor of $\gamma = 1.2$ expands the spectral dynamic range. It attenuates the lower-magnitude values while leaving the peaks relatively unchanged. Audibly, this silences background noise and amplifies the most dominant frequencies. On the spectrogram, this expansion creates a much darker image.

Part 3: AM vs FM Modulation

The spectra of Amplitude Modulated (AM) and Frequency Modulated (FM) signals were compared. The AM spectrum contained only three distinct peaks: the carrier frequency and two sidebands. In contrast, the FM spectrum was significantly wider. This is because AM simply changes the amplitude of the carrier wave, whereas FM constantly varies the frequency of the carrier wave which produces a wide spread of multiple sidebands.

Part 4: Voiced vs Unvoiced Features (Energy and ZCR)

Normalized Energy and Zero-Crossing Rate (ZCR) were extracted to classify phonemes in a speech signal.

Voiced phonemes are produced by vocal cord vibrations, which generate high-amplitude, low-frequency periodic waveforms. Therefore, voiced regions are characterized by high energy and low ZCR. Unvoiced phonemes (for example, s or f) are produced by turbulent airflow, resembling high-frequency noise. Consequently, unvoiced regions are identified by low energy and high ZCR.

Part 5: Fourier Uncertainty

A Gaussian wave packet was analyzed in both the position domain (x) and the momentum domain (k) to explore Fourier uncertainty.

As σ_x decreases, the k -space Δk increases. The relationship is strictly inverse; a highly localized position yields a broad frequency spectrum.

For all evaluated Gaussian cases, the product of the uncertainties $\Delta x \Delta k$ is exactly 0.5. This represents the theoretical minimum bound for Fourier transform pairs.

Using the relation $p = \hbar k$ (with $\hbar = 1$), the momentum uncertainty is $\Delta p = \Delta k$. Thus, the product evaluates to $\Delta x \Delta p = 0.5$, which satisfies the inequality bound $\Delta x \Delta p \geq \hbar/2$.

This mathematical property of the Fourier transform is the example of Heisenberg's Uncertainty Principle. Because position (x) and momentum (p) are conjugate variables related by a Fourier transform, a state cannot be infinitely localized in both domains simultaneously. Sharpening position inherently broadens momentum.